

Python Practice: Building Simple Programs

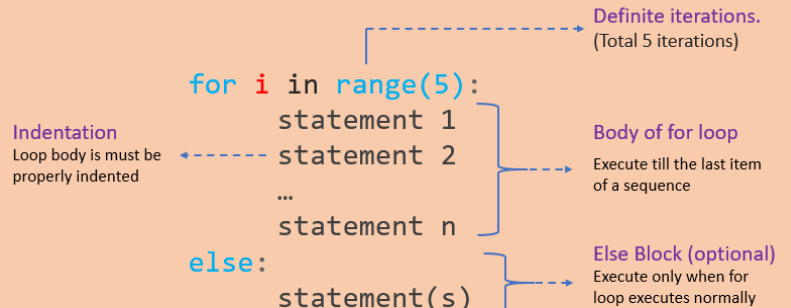
In this worksheet, you will start writing longer pieces of code that behave like small programs.

Each section builds on the previous one. You will combine variables, loops, conditionals, and functions to make code that reacts, repeats, and changes based on input.

Run every example before changing it.

Python for loop

A for loop is used for **iterating over a sequence and iterables** (like range, list, a tuple, a dictionary, a set, or a string).



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Section 1: Counting With Rules

Task 1

Run this code

```
for i in range(1, 11):  
    print(i)
```

Change the code so it prints only numbers from 1 to 5

Task 2

Run this code

```
for i in range(1, 11):  
    if i > 5:  
        print(i)
```

Write down what it prints

Task 3

Change the code so that it prints only numbers **less than 5**

Section 2: Keeping Score

Task 4

Run this code

```
score = 0

score = score + 1
print(score)

score = score + 1
print(score)
```

Write down what it prints

Task 5

Run this code

```
score = 0

for i in range(5):
    score = score + 2

print(score)
```

Change the code so the final score is 15

Task 6

Run this code

```
score = 10

if score >= 10:
    print("Level up!")
else:
    print("Keep trying")
```

Change the score so that it prints Keep trying

Section 3: Working With Messages

Task 7

Run this code

```
message = "HELLO"

for letter in message:
    print(letter)
```

Change the message to your own word

Task 8

Run this code

```
message = "HELLO"

for letter in message:
    print(letter + "!")
```

Change the code so that it prints:

H?

E?

L?

L?

O?

Task 9

Run this code

```
message = "PYTHON"

for letter in message:
    if letter == "O":
        print("0")
    else:
        print(letter)
```

Write down what it prints

Task 10

Change the code so that:

- A becomes @

- Everything else stays the same

Section 4: Functions That Do More

Task 11

Run this code

```
def print_score(score):  
    if score >= 10:  
        print("High score!")  
    else:  
        print("Score:", score)  
  
print_score(12)
```

Change the function call so that it prints Score: 5

Task 12

Run this code

```
def repeat_message(message, times):  
    for i in range(times):  
        print(message)  
  
repeat_message("Hi", 3)
```

Change the function call so that it prints your name 6 times

Section 5: Making a Mini Program

Task 13

Run this code

```
def countdown(start):  
    for i in range(start, 0, -1):  
        print(i)  
    print("GO!")  
  
countdown(3)
```

Change the countdown so that it starts at 5

Task 14

Run this code

```
def check_password(password):  
    if password == "python":  
        print("Access granted")  
    else:  
        print("Access denied")  
  
check_password("python")
```

Change the function call so that it prints Access denied

Section 6: A Longer Example (Read Carefully)

Task 15

Run this code

```
def game_score(points):  
    score = 0  
  
    for p in points:  
        if p == "W":  
            score = score + 3  
        elif p == "D":  
            score = score + 1  
        else:  
            score = score + 0  
  
    print("Final score:", score)
```

Write down what it prints

Task 16

Change the string so that the final score is 8

Section 7: Mini Challenge

Task 17

Run this code

```
def censor(message):  
    result = ""  
  
    for letter in message:  
        if letter == "A":  
            result = result + "*"  
        else:  
            result = result + letter  
  
    print(result)  
  
censor("BANANA")
```

Write down what it prints

Task 18

Change the code so that:

- All vowels (A, E, I, O, U) become *
- Everything else stays the same

Final Notes

You are now writing longer code that:

- Keeps track of values
- Uses rules to make decisions
- Changes text
- Acts like a real program

These are the same skills used in:

- Games
- Message ciphers
- Graphics (like turtle)
- Real-world programs